

EARLIER VERSIONS OF WINDOWS.

WINDOWS 3.1

Windows 3.1 is not an OS. In order to install and run Windows 3.1, it requires one to have DOS already installed on the computer.

Windows 3.1 simply replaces the DOS interface, & changes the way in which the user communicates with the computer, gives commands, and works with the application programs.

Since Windows 3.1 is not an OS, it does not perform the function of resource management. The Processor, Input/Output device & Disk management is handled by DOS. However, Memory management & Disk-caching are performed by Windows to improve the performance of Windows applications and to ensure efficient multi-tasking.

Windows for Workgroups (WINDOWS 3.11).

Windows 3.11 adds Networking capabilities & additional features to Windows 3.1.

With Windows 3.11, you can connect to several PCs into a Peer-to-Peer network & be able to share resources (Files, Printers, Modems, etc), connect to other networks, and also send & receive E-mail and Faxes.

Windows 3.11 has network programs such as, *NetWatcher* & *Winmeter* that enables one to monitor the performance of the system & also how other people are using the shared resources. *Chat* program allows one to communicate with other workgroup members by sending & receiving messages through typing.

CURRENT VERSIONS OF WINDOWS.

WINDOWS '95.

Windows 95 is currently the defacto (actual / effective) OS for IBM compatible PCs. It came as an enhancement of Windows 3.11.

It is a true 32-bit OS that manages the entire computer system, including its Screen, Keyboard, Disk drives, Memory & the Processor (i.e. it does not run on top of DOS).

Windows 95 offers high performance & improvements to many familiar features in the earlier versions of Windows.

- (i). It is compatible with the previous DOS & Windows applications enabling those applications to run side by side with the newer 32-bit Windows 95 applications.
- (ii). **New User Interface.**
 - Has a *Start button* that provides easy access to programs, most recently used documents, and important system utilities such as Control Panel & Printers.
 - Easier to switch between programs using the Taskbar.
 - The look of the maximize & minimize buttons has been changed and a Close button has been added.
- (iii). **Ability to handle Long File Names.**

Win '95 supports long file names; up to 32 characters, thus allowing the user to give files more meaningful names. This also makes the files easier to find and organize.
- (iv). **Built-in Networking Support.**

Win '95 supports a wider choice of networks & has its own suite of transmission protocols. It has facilities for sharing files & devices, which are completely integrated in the Windows 95 user interface.
- (v). **Integrated Mail and Fax support.**

Win '95 includes *Microsoft fax*. It also includes *Microsoft Exchange*, which can be used to view & work with all types of electronic communications, including E-mail.

(vi). Support for Mobile Computing.

Win '95 supports Mobile computing with a file synchronization tool & a utility for transferring files over a direct cable link or dial-up networking.

(vii). Win '95 allows true *Multi-tasking*, i.e. one can have several processes running in the background while he/she works.

(viii). Improved Game & Multimedia Support.

Win '95 has inbuilt sound, video & CD capabilities, and can fully support MS-DOS based games. It also has improved performance for playing video & sound files.

(ix). Plug-and-Play Hardware compatibility.

Win '95 supports Plug-and-Play devices, which makes it easier to install new devices onto the system.

WINDOWS '98.

Win '98 is an enhancement of Win '95. It is similar to Windows 95 & all applications written for Win '95 are able to run without any modification under Windows 98. However, Windows 98 doesn't 'crash' as often as Win '95.

Win '98 is able to support the latest graphics, sound & multimedia technologies. It enables easy addition & removal of peripheral devices with support for the *Universal Serial Bus* (USB). It also supports Web TV.

Win '98 has a better interface to the Internet. The **Active Desktop** lets you place the Channel Bar & Internet components such as regularly updated news headlines right on the Windows desktop. Windows 98 even lets you set a Web Site as the desktop's background paper.

NEW FEATURES IN WINDOWS '98.

1. Improved Web features.

Windows 98 combines the power of the computer with the interactive content of the Internet thus improving the Web features:

-)] **AutoComplete** feature, which automatically completes previously visited Web addresses as you type them.
-)] Has an improved listing of favorite Web sites.
-)] Has improved tracking of Web sites visited.
-)] Supports all major Internet standards, including **ActiveX**, **Java**, etc.
-)] Improved performance with Dynamic HTML, which makes Web pages more interesting.

New Internet support.

The *Internet Connection wizard* helps you sign up for Internet service through an **Internet Service Provider** (ISP) & automatically handles the software configuration steps necessary for gaining access to the Internet.

Active Desktop for displaying Web pages on the Desktop.

Active Desktop makes it possible for you to customize your desktop, launch programs, switch between files & keep up with the latest World news by integrating the Web and your desktop. With Active Desktop, you can update your Web elements at any time.

To make a Web-page picture your desktop wallpaper.

1. Right-click the desktop, then click **Properties**.
2. Open the **Background** tab, then click **Browse**.
3. Click the HTML document you want, then choose **Apply**.

The wallpaper appears on your desktop as tiled, stretched, or centered, depending on the current setting in the **Display** box.

Tip. You can also right-click any graphic on a Web page, then click **Set as Wallpaper**.

Browse the Web from anywhere.

The **Go & Favorites** menus, which help you reach your favorite Web sites are found everywhere in Windows 98. E.g., they can be found on the **Start** menu or the **Internet Explorer** icon on your Taskbar.

You can use the Address bar in Windows Explorer or add an *Address bar* to the Taskbar, and type an Internet address without first opening the browser.

Microsoft Outlook Express for E-mail.

Win '98 includes **Microsoft Outlook Express**, which provides features for e-mail and newsgroup communications.

To start Outlook Express, click **Start**, point to **Programs**, then click **Outlook Express**.

NetMeeting & Microsoft Chat for Internet Chatting.

Win '98 includes *NetMeeting* that helps one to hold digital conversations with family, friends, and business associates around the world. It can also assist you to work with a group of people from within any Windows-based program by sending text messages, and transferring files.

With the correct hardware installed on your computer, you can also have live, face-to-face meetings.

Microsoft FrontPage Express for creating Web Pages.

FrontPage Express is a Web-page editor that helps you to create simple web pages & also copy your web pages to a web server.

To start FrontPage Express, click **Start**, point to **Programs**, point to **Internet Explorer**, then click **FrontPage Express**.

Power management.

Power Management helps to save power resources by putting the computer on Standby or Hibernation.

Power Management works on computers with **Advanced Power Management (APM)**, and on computers that use the **Advanced Configuration and Power Interface (ACPI)**.

Universal Serial Bus (USB).

The **Universal Serial Bus** makes the computer easier to use with advanced plug-and-play capabilities. Using the USB connector, you can add devices to your computer easily without having to restart.

Accessibility wizard

The *Accessibility wizard* makes it easier for people with disabilities to operate a computer without installing special software.

Accessibility options—such as StickyKeys, ShowSounds, and MouseKeys—are designed to help users with specific disabilities make full use of the computer.

- To start the Accessibility wizard;

Click **Start**, point to **Programs**, point to **Accessories**, point to **Accessibility**, then click **Accessibility Wizard**.

Help

Win '98 has an Help system that makes a computer easier to use. With Help, you can find answers to your questions quickly & easily. You can also get up-to-date technical support from the World Wide Web.

To start Help, click **Start**, then click **Help**. To get help from the Web, click **Web Help** at the top of the Help page.

2. Improved reliability

Windows 98 improves computer reliability by introducing new wizards, utilities, and resources such as, **Windows Update** that help keep your system running smoothly.

Windows Update is a Web-based (online extension) of Windows 98 that helps your computer work & run better.

Windows Update provides a central location to find and download product enhancements such as system files, device drivers, new Windows 98 features and specific files customized for your computer.

- To open Windows Update, click **Start**, then choose **Windows Update**.

-Or-

Click **Start**, point to **Settings**, then click **Windows Update**.

To add or change a device driver for hardware.

1. Open **Control Panel**, then double-click **System** to open the **System Properties** dialog box.
2. Click the plus sign (+) next to the type of hardware, and then double-click the hardware.
3. On the **Driver** tab, click **Update Driver**, then follow the instructions on the screen.

Note. Some devices do not have a **Driver** tab. In this case, change the driver by double-clicking the icon in **Control Panel** that is specific to that type of hardware.

Putting toolbars where you want them.

You can add toolbars to your Windows taskbar, making it even easier to get to your programs, files, folders, and favorite Web pages.

The following toolbars can be added.

- (i). **Quick Launch** toolbar - provides shortcuts to several often-used features, including the desktop, the Web browser, and Outlook Express.
- (ii). **Address** toolbar - enables one to type a Web page address (URL) without first opening the Internet Explorer browser.
- (iii). **Links** toolbar - provides shortcuts to important Web sites, e.g., www.Microsoft.com, so that one can open them without first opening the browser.
- (iv). **Desktop** toolbar - contains all of the shortcuts on your desktop and organizes them into one convenient place.

To add a toolbar to the taskbar.

1. Right-click a blank area on the Taskbar, point to **Toolbars**, then click **Address**, **Links**, **Desktop**, or **Quick Launch**.

The selected toolbar appears on the **Taskbar**.

3. A faster Operating system.

Win '98 includes a suite of programs that help your computer run faster than Windows 95 without adding new hardware.

E.g., System tools like,

- **Maintenance wizard**, which schedules Windows' housekeeping programs such as *Scandisk*, *Disk Cleanup*, & *Disk Defragmenter* to run on regular basis.
- **Drive Converter (FAT32)**.

4. More entertaining and fun.

Win '98 makes your computer more entertaining by introducing new features such as enhanced television, video playback, and support for new hardware.

- (a). **New hardware and graphics**

Windows 98 supports a variety of new hardware devices, such as **USB & IEEE 1394** devices, *DVDs, force-feedback Joysticks, digital audio Speakers, recording devices* and improved plug-and-play capabilities that make installing new hardware easier.

(b). DirectX

Win '98 includes **DirectX**, a utility that provides better playback of different types of Multimedia & manages 3-D graphics on your computer.

(c). Desktop themes

Win '98 includes 4 Microsoft themes—Underwater, Jungle, Baseball, & Space and the entire Microsoft Plus! for Windows 95 themes package.

WINDOWS Me (Millennium Edition).

Windows Me creates a mainstream OS that incorporates the best business features of Win '98 & builds on the strengths of Windows NT.

The in-built features of Windows 98 include: - a simplified user interface, improved Plug-and-Play, Power management, & support for a broad range of hardware devices.

Windows Me also builds on the strengths of Windows NT Workstation. This makes Windows Me secure, reliable, perform better, and manageable.

OS/2.

OS/2 is IBM's GUI Operating System for Intel 8086 microprocessors. It is a 32-bit Single-user, Multi-tasking OS.

OS/2 cannot support terminals since it is Single-user.

OS/2 is more robust & faster than Windows 3.1 & Windows 3.11. When Multi-tasking under OS/2, the computer does not hang-up when one of the applications fails. However, the applications running on it are few and therefore, applications portability is a problem.

NETWORK OPERATING SYSTEMS.

A **Computer Network** is formed whenever more than one computer is connected together.

A computer network usually has one computer reserved as the **Server** or "Mother" of all the other computers on the network. The network Operating system usually resides in the Server.

Network Operating systems consist of several pieces of software that run in various devices connected to a computer network.

A good network OS is required to provide the following services: -

- ✓ Client-Server capabilities.
- ✓ File sharing services.
- ✓ File/Directory Naming service, i.e., should provide efficient directory naming in the network.
- ✓ Print sharing services, e.g., print spooling and queue management.
- ✓ Mail services.
- ✓ Network management services, such as, setting up of users, adding resources & managing network failures.

Advantages of Networking computers.

- (i).** *Sharing of file servers* – files are stored in a centralized file server, making it possible to control who has access to which files & the nature of the access.
- (ii).** *Sharing of Printers.* It enables several users to print using one Printer. Jobs can be stored in the print server until the printer is ready to accept the information. This reduces the user's time wasted waiting for the printer.
- (iii).** *Sharing of powerful computers.* It is possible to harness the computational power of a more powerful computer remotely.

- (iv). *Sharing of communication ports.* It provides access to shared communications ports such as, Modems.
- (v). *Centralized backup.* It solves the problem of maintaining regular backup of many computers by backing up only the file server.
- (vi). *Electronic mail (e-mail).* Networks enhance communication between people by use of fast electronic mails.

Examples Network operating systems.

WINDOWS NT.

Windows NT (**New Technology**) is a true 32-bit OS, designed to run on Intel-based 32-bit microprocessors.

There are 2 versions of Windows NT: -

) **Windows NT Advanced Server.**

) **Windows NT Workstation.** Windows NT Workstation is similar to Windows 95/98.

Windows NT is technically superior to Windows 95/98. For instance, it is more secure and keeps buggy programs from crashing the whole PC.

Windows NT is written to operate on both DOS & UNIX platforms, allowing applications to execute on different types of computers.

With Windows NT, Networking is an integral part of the OS. Network management functions are generally simplified & easier to perform.

Drawbacks of Windows NT.

- (i). It has limited hardware & software compatibility.
- (ii). Requires its own drivers for devices, such as Video cards, Printers, etc. This is because, the existing DOS, Windows 3.1, & Windows 95 drivers won't be able to work.

NOVELL NETWARE.

It a *Local Area Network (LAN)* OS designed for IBM standard computers running MS-DOS, Windows, OS/2 or Macintosh computers.

Novell Netware provides a means for connecting several computers together into computer network.

It also gives all the computers on the network, the ability to access files, programs, printers, e-mail & other services (collectively called network **Resources**) that are attached to the network.

In addition, Novell Netware has many features for security, backup, and network administration.

The later versions of Netware offer connectivity options for a wide range of Mainframe and Minicomputers & support Wide Area Networking (WAN). This makes it possible to have large corporate networks spanning the globe.

UNIX.

UNIX was developed by the Bell Laboratories in U.S.A.

UNIX belongs to the **MATS** (Multi Access Time-sharing Systems) family of Operating systems.

Features.

UNIX is a full Multi-tasking OS. **Multi-tasking** means that, the system can perform several tasks (called *Processes*) at the same time. This enables the user to give the system one or more tasks to perform "in the background" while he/she gets on with something else without having to wait for the task to finish.

It is a Multi-user Time-sharing OS written mostly in **C**, making it easily portable between machines with different hardware facilities. **Multi-user** means that, more than one user can use the system at the same time.

Has a sophisticated and powerful command interpreter called **SHELL**.

UNIX is now widely used, as it has become the OS that represents the link between the majority of Mainframes, Minis, Workstations, PCs, Supercomputers and Networks.

LINUX.

Linux is a UNIX-type Operating system.

It includes true Multitasking, Virtual memory, shared libraries, proper memory management and other features consistent with UNIX-type systems.